# HARSH TOMAR

AI / GenAI Engineer

+91-8817969476 • tomarharsh28303@gmail.com • LinkedIn • GitHub • Portfolio

## SUMMARY

AI engineer building **LLM-powered systems**: RAG pipelines, LangGraph agents, LLMOps, and domain-specific fine-tuning. Built **Continuum**, an LLMOps platform for LLM quality monitoring on AWS; built and deployed a RAG service at **i3 Digital Health**; built **QuantaAI**, a LangGraph multi-agent chatbot with autonomous web-search routing; fine-tuned **Qwen 2.5 1.5B** for Medical Q&A via QLoRA (**ClinicalQwen**); and reimplemented **PaLiGemma VLM** in PyTorch for architectural depth. Hands-on with LangChain, LlamaIndex, vector DBs (Pinecone, Chroma, FAISS, Weaviate), and OpenAI / Anthropic / Gemini APIs.

## EXPERIENCE

**AI Intern** **May 2025 – Jul 2025**

*i3 Digital Health* *Remote*

- Built and deployed a **RAG pipeline for researcher profiling**, aggregating and cleaning data from PubMed, Google Scholar, and ResearchGate via multi-source APIs and scraping; **LangChain** retrieval over vector embeddings drove contextual collaborator recommendations
- Owned the full build-to-deploy loop: iterated chunking strategies, embedding models, and retrieval prompts with healthcare SMEs to improve retrieval quality; containerised and deployed the service via **FastAPI + Docker**
- Worked directly with domain experts in an unfamiliar healthcare domain, translating ambiguous requirements into a working retrieval system within the internship window

## PROJECTS

**Continuum – LLMOps Platform for Quality Monitoring & Self-Healing** **2026**

*AWS ECS Fargate, Apache Kafka, FastAPI, React/TypeScript, Terraform, PEFT, MLflow, Prometheus* | GitHub

- **Problem:** detect LLM quality degradation in production traffic and orchestrate fine-tuning to remediate it
- **Approach: 11-microservice event-driven platform** on **AWS ECS Fargate** with an **Apache Kafka** trace pipeline; multi-dimensional quality scoring (Claude LLM-judge + ROUGE/BLEU + semantic similarity); **Statistical Process Control** (Western Electric rules) for drift detection; designed self-healing loop for **LoRA** fine-tuning with A/B canary deployment; full **Terraform IaC** and GitHub Actions CI/CD
- **Result:** verified end-to-end trace flow (Kafka → scoring → PostgreSQL) with **166 Prometheus metric series**, an authenticated REST API, a React dashboard, and 10/10 passing E2E tests

**QuantaAI – LangGraph Search-Routing Chatbot** **2025**

*LangGraph, GPT-4, Tavily Search, FastAPI, Next.js 14, TypeScript* | GitHub

- **Problem:** chat assistant that decides on its own when a query needs live web search vs. direct generation
- **Approach: 4-node LangGraph state machine** (Classifier → Search → Generate → Response) with a GPT-4 routing classifier; **Tavily** web-search integration with relevance ranking; SSE streaming via FastAPI to a Next.js 14 frontend; sliding 10-message context window
- **Result: 94% routing-classifier accuracy** on a hand-labeled eval set; transparent multi-stage UX exposing classification, search, and generation phases

**VLMverse – PaLiGemma Vision-Language Model** **2025**

*PyTorch, SigLIP, Gemma 2B, RoPE, KV-Cache, Grouped-Query Attention, RMSNorm* | GitHub

- **Problem:** internalise modern VLM architecture by reimplementing one end-to-end
- **Approach:** reimplemented **PaLiGemma** in pure PyTorch following the paper + reference walkthroughs — SigLIP encoder (16×16 patches → 196 tokens), **Gemma-2B** decoder, **RoPE** positional embeddings, **KV-Cache** for autoregressive inference, grouped-query attention, RMSNorm
- **Result:** working end-to-end inference; deep working knowledge of attention, positional encodings, and cross-modal projection

**ClinicalQwen – Medical Q&A Fine-Tuning on Qwen 2.5 1.5B** **2026**

*PyTorch, Qwen 2.5 1.5B, QLoRA, 4-bit NF4, PEFT, Hugging Face* | GitHub

- **Problem:** adapt a general-purpose LLM to answer clinical and medical queries accurately
- **Approach: QLoRA** fine-tuning of **Qwen 2.5 1.5B** on **5,000 medical Q&A samples** (lavita/medical-qa-datasets); 4-bit NF4 quantisation; PEFT with LoRA rank-8 adapters; trained on Google Colab T4
- **Result:** domain-adapted medical LLM with ∼1% trainable parameters (9.2M / 1,552M); 86% keyword coverage, perplexity 3.14; deployed on Hugging Face Hub

**rPPG Heart-Rate – Contactless Biosignal Extraction** **2026**

*PyTorch, PhysNet, ONNX, FastAPI, OpenCV, Signal Processing* | GitHub

- **Problem:** extract heart rate from subtle skin-colour variations in webcam video — no physical sensors required
- **Approach:** hybrid pipeline combining classical algorithms (CHROM, POS) with **PhysNet** deep learning model trained on UBFC-rPPG dataset; **ONNX export** for ∼12 FPS real-time CPU inference; **FastAPI** web dashboard with per-method confidence scoring
- **Result:** contactless real-time heart-rate estimation deployable on CPU without GPU dependency

**Breast Cancer Histopathology Analysis** **2025**

*PyTorch, MobileNetV2, Grad-CAM, OpenCV, Transfer Learning, Medical AI* | GitHub

- **Problem:** automated cancer detection from histopathology images with explainable predictions for clinicians
- **Approach: MobileNetV2** transfer learning for benign/malignant classification; **Grad-CAM** heatmaps highlighting decision regions; OpenCV-based nuclear density and chromatin pattern extraction
- **Result:** automated clinical report generation with confidence scores, heatmap overlays, and feature analysis

**Additional Work**

**IntelliMail** — AI agent for hyper-personalised course recommendation emails [GitHub] • **RAGCraft** — hub for RAG patterns: advanced chunking, hybrid search, re-ranking [GitHub] • **100-Days of Inference Engineering** — daily LLM inference experiments and visualisations [GitHub] • **LoRA / QLoRA** — pure PyTorch LoRA ($W = W_0 + BA$, rank 8/16/32) + QLoRA with 4-bit NF4; ∼65% memory reduction on BERT • **Transformers-CV** (**173 commits**) — Flamingo VLM, DDPM, ViT, JEPA, VAE/VQ-VAE [GitHub] • **Tennis Vision** (**35 stars**) — 30 FPS sports CV pipeline: YOLOv8x ball detection + ResNet-50 court-keypoint regression, deployed on HF Spaces [GitHub]

## Open Source Contributions

**3 merged PRs to upstream projects:** BBoxMaskPose #12 (Docker support, ICCV 2025 paper repo) • multimodal-agents-course #23 (Kubrick UI build fix) • hive #6849 (deprecated storage refactor)

## Technical Skills

**LLM & GenAI:** LangChain, LangGraph, LlamaIndex, RAG, prompt engineering, agent orchestration, OpenAI / Anthropic / Gemini APIs, Hugging Face Transformers
**Fine-Tuning & Architecture:** LoRA, QLoRA, PEFT, 4-bit NF4 quantisation, attention mechanisms, RoPE, KV-Cache, grouped-query attention, RMSNorm
**Vector & Retrieval:** Pinecone, Chroma, FAISS, embedding models, semantic search, chunking strategies, re-ranking
**Deployment & LLMOps:** FastAPI, Docker, Kubernetes, AWS (ECS Fargate, EKS, EC2, S3, ECR), Apache Kafka, Terraform, MLflow, Prometheus, Grafana, GitHub Actions CI/CD, Hugging Face Spaces
**Foundations:** PyTorch (primary), TensorFlow, NumPy, Pandas, transformers, autoregressive decoding |
**Languages:** Python, SQL, TypeScript, Bash

## Education

**B.Tech in Artificial Intelligence & Data Science** **Nov 2022 – May 2026**
Lakshmi Narain College of Technology, Bhopal CGPA: 7.2/10
*Coursework:* NLP, ML, DL, Reinforcement Learning, Linear Algebra, Probability & Statistics, Neural Networks